SPINNAKER ANALYTICS

Classifying Shooting Incident Fatality

Using Machine Learning

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AGENDA

- Introduction and Problem Statement
- Objectives and Dataset Overview
- Exploratory Data Analysis (EDA)
- Preprocessing
- ML Models Implementation
- Results and Performance Metrics
- Key Insights, Applications, Conclusion and Future Work

Introduction

Shooting incidents are serious crimes that impact public safety and communities. Analyzing them helps identify patterns and high-risk areas, enabling better prevention and resource allocation. This project uses machine learning to analyze crime data and identify patterns for accurate classification. By leveraging location, time, and demographic features, it enables better insights and supports improved public safety decisions.

Problem Statement

Shooting incidents pose a major threat to public safety and are influenced by factors such as location, time, and demographics. The challenge is to accurately analyze these factors and build a model that can classify incidents effectively. Additionally, the presence of imbalanced data makes it difficult for traditional models to correctly predict minority cases, requiring advanced techniques for better performance.

Objectives

- Analyze dataset
- Perform preprocessing
- Handle imbalance using SMOTE
- Apply ML models
- Compare performance

Dataset Overview

The dataset consists of 27,312 rows and 21 columns before performing Exploratory Data Analysis (EDA), containing features related to location, time, and demographics of shooting incidents.

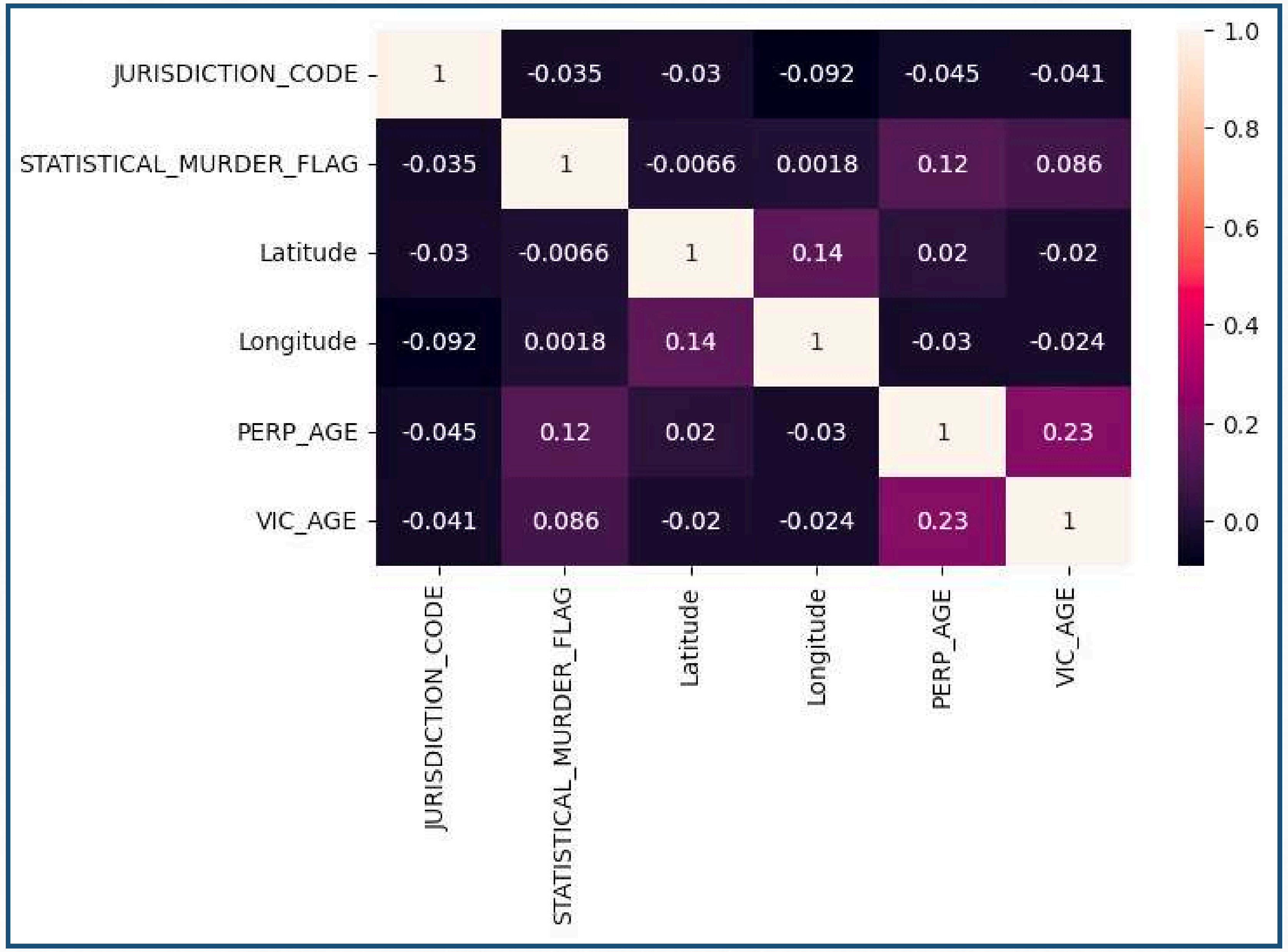
BORO	PRECINCT	JURISDICT	STATISTIC	VIC_SEX	VIC_RACE	Latitude	Longitude	VIC_AGE
QUEENS	105	0	FALSE	M	BLACK	40.66296	-73.7308	21
BRONX	40	0	FALSE	M	BLACK	40.81035	-73.9249	21
QUEENS	108	0	TRUE	M	WHITE	40.74261	-73.9155	35
BRONX	44	0	FALSE	M	WHITE HIS	40.83778	-73.9195	17
BRONX	47	0	TRUE	M	BLACK	40.88624	-73.8529	55
BROOKLYI	81	0	TRUE	M	BLACK	40.67846	-73.928	35
QUEENS	114	0	FALSE	M	BLACK	40.75648	-73.9473	35
BROOKLYI	81	0	TRUE	M	BLACK	40.69426	-73.9328	21
QUEENS	105	0	FALSE	M	BLACK	40.70611	-73.7471	35
QUEENS	101	0	FALSE	M	BLACK	40.5977	-73.7491	35
MANHATT	23	2	FALSE	M	BLACK	40.79773	-73.9465	35
BROOKLYI	75	0	FALSE	M	BLACK	40.66061	-73.8959	35
BROOKLYI	71	0	FALSE	M	BLACK	40.66502	-73.9571	35
BRONX	50	0	FALSE	M	WHITE	40.88449	-73.9056	21
BROOKLYI	78	2	FALSE	M	BLACK HIS	40.68257	-73.985	35

Exploratory Data Analysis (EDA)

- Exploratory Data Analysis (EDA) was performed to understand the structure and patterns in the dataset.
- Various visualizations were used to analyze the distribution of incidents, trends over time, and relationships between features such as age, gender, and location.
- The analysis also helped identify class imbalance, missing values, and key factors influencing shooting incidents.

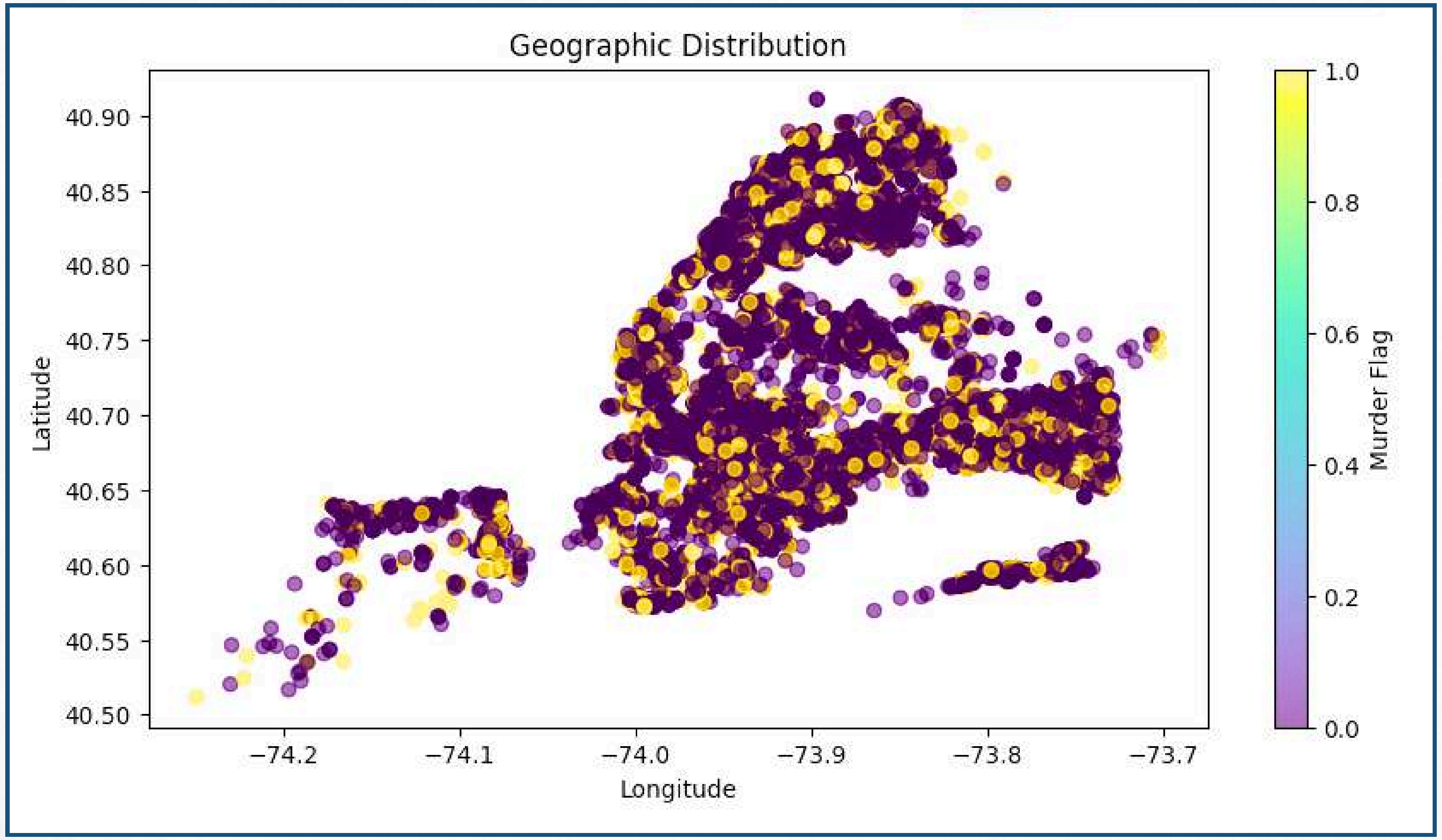
1. Correlation Heatmap :-

The correlation heatmap shows that most features have weak relationships, with only a moderate negative correlation between precinct and latitude, indicating minimal linear dependency overall in the dataset.



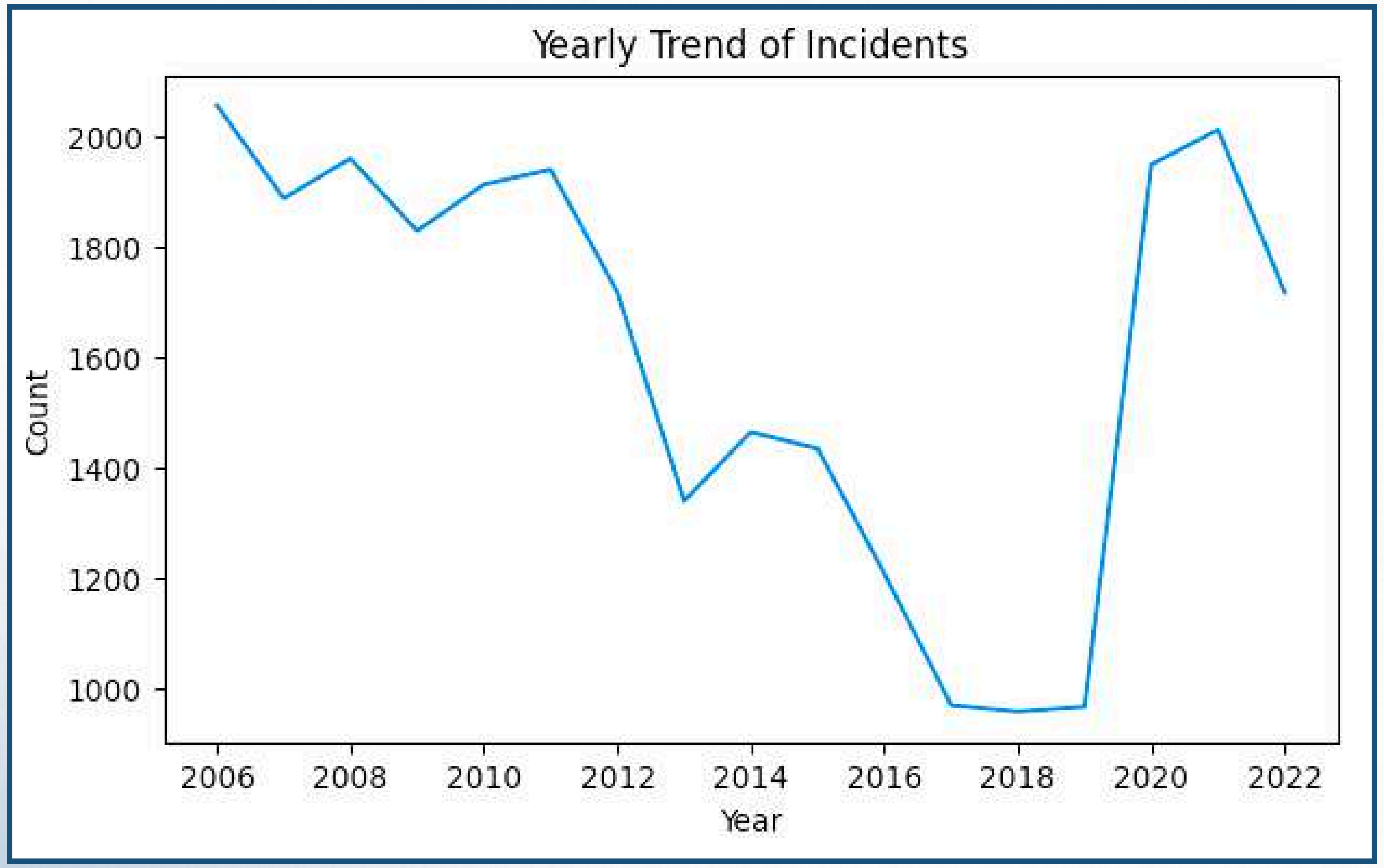
2. Geographic Distribution :-

Incidents are geographically clustered in specific regions, indicating hotspot areas rather than uniform distribution.



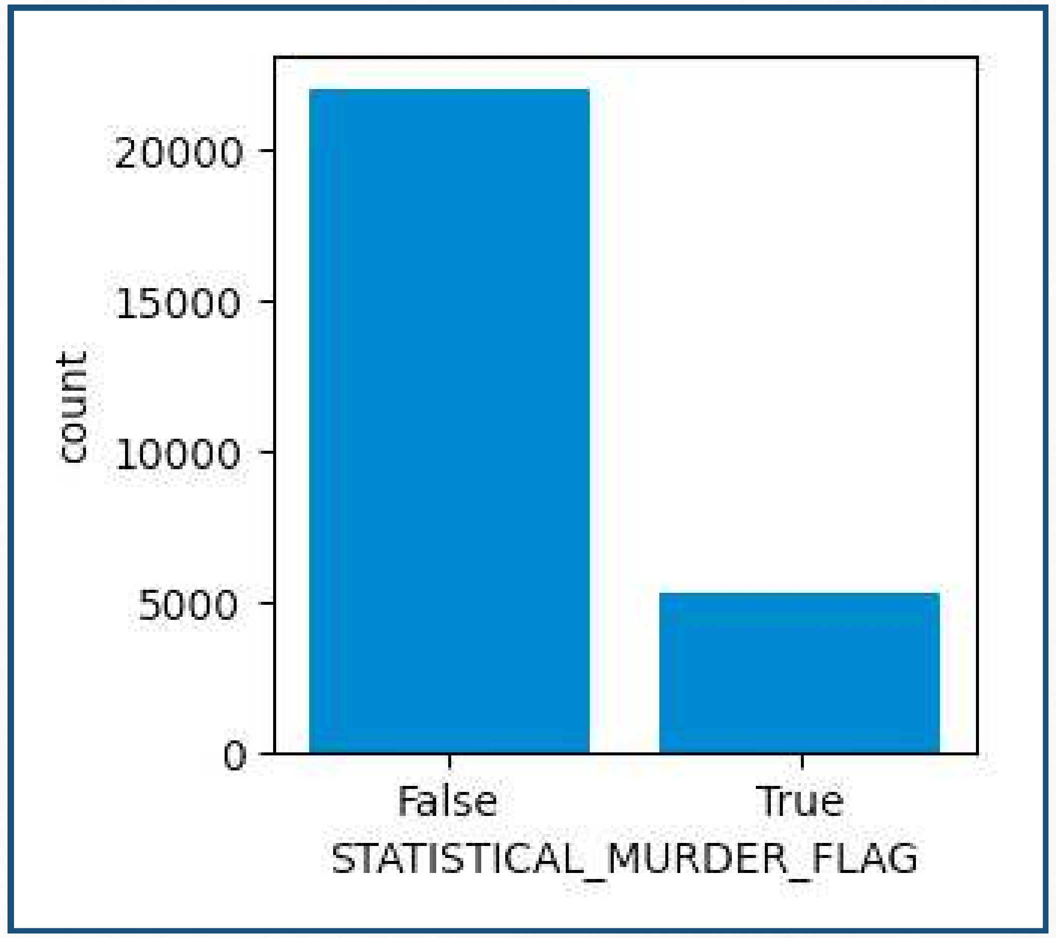
3. Yearly Trend of Incidents :-

The line plot illustrates the yearly trend of incidents, showing how the number of cases changes over time and highlighting any increasing or decreasing patterns.



4. Class Imbalance :-

The dataset is imbalanced, with non-fatal cases far exceeding fatal ones. This can bias the model toward the majority class, making it harder to predict minority cases, hence techniques like SMOTE are needed.



Preprocessing

Data Cleaning :-

- Dropped columns with high percentage of missing values.
- Filled missing categorical values.
- Divided data into 70% training and 30% testing.

Feature Engineering :-

- Extracted the Year, Month and Day from OCCUR_DATE as it contained mixed date format.

Encoding Techniques :-

- One-Hot Encoding – Applied on BORO, PERP_SEX, PERP_RACE, VIC_SEX, VIC_RACE due to low cardinality.

ML Models Impelementation

1. Logistic Regression :-

   Logistic Regression was included as a simple and interpretable starting point for murder flag classification.

2. Random Forest :-

   Random Forest was chosen for its ability to model non-linear patterns and it works well with imbalanced data.

3. XGBoost :-

   XGBoost was applied due to its high accuracy, fast computation, and effectiveness on complex data patterns.

Hyperparameter Tuning :-

Utilized GridSearchCV for both Random Forest and XGBoost to optimize parameters.

Results & Performance Metrics

- The models were evaluated using metrics such as accuracy, precision, recall, and F1-score.
- Before applying SMOTE, the models showed high accuracy but poor recall for the minority class.
- After applying SMOTE, there was a significant improvement in recall and F1-score, with Random Forest and XGBoost achieving the best overall performance (~85% accuracy) and better class balance.

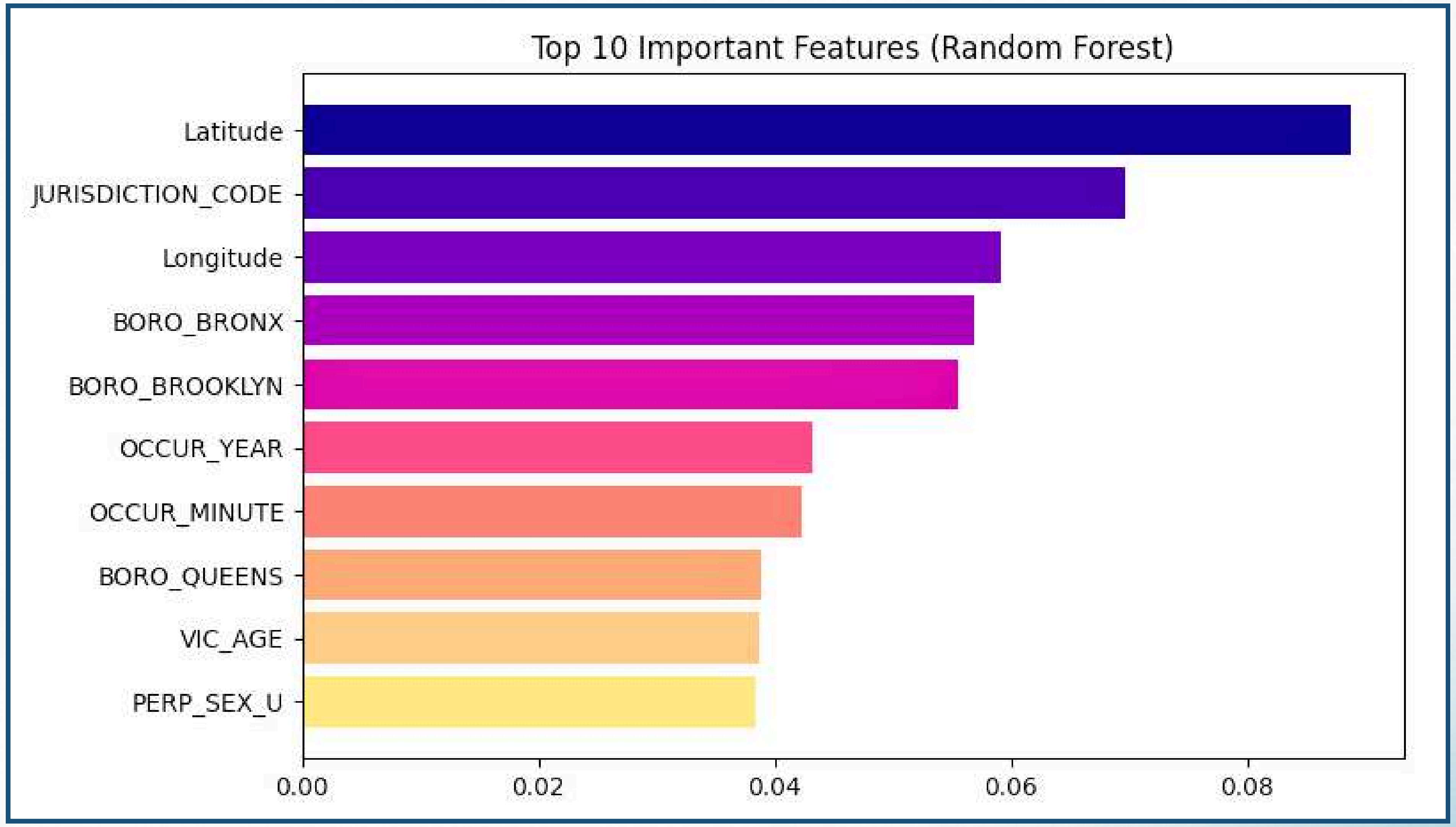
Model	**Accuracy**	**Precision**	**Recall**	**F1-score**
Logistic Regression	67.79	0.52	0.52	0.52
Random Forest	85.05	0.87	0.85	0.85
XGBoost	85.16	0.87	0.85	0.85

Feature Importance

- The model is primarily driven by location-based features, with Latitude and Longitude having the highest importance, indicating that geographical patterns strongly influence predictions.

- Administrative and regional variables like JURISDICTION_CODE and borough features also contribute significantly.

- Temporal (OCCUR_YEAR, OCCUR_MINUTE) and demographic (VIC_AGE, PERP_SEX) features have comparatively lower but still meaningful impact.



Key Insights

- Location strongly impacts prediction.
- Class imbalance affects model performance.
- SMOTE improves results.
- Ensemble models perform best.

Applications

- Crime prediction systems
- Resource allocation
- Risk analysis
- Public safety

Conclusion

- SMOTE improved model performance.
- Random Forest & XGBoost performed best.
- Importance of handling imbalance.

Future Work

- Use more data
- Try deep learning models
- Deploy real-time system

THANK YOU